Exercise 3 – Classification I

Introduction to Machine Learning

Exercise 1: Logistic regression – softmax link and decision boundary

The whole exercise is only for lecture group A!

Learning goals

1. Solve "show equivalence"-type questions
2. Derive the decision boundary of the logistic classifier

Binary logistic regression is a special case of multiclass logistic, or softmax, regression. The softmax function is the multiclass analogue to the logistic function, transforming scores $\boldsymbol{\theta}^\top \mathbf{x}$ to values in the range [0, 1] that sum to one. The softmax function is defined as:

$$\pi_k(\mathbf{x}|\boldsymbol{\theta}) = \frac{\exp(\boldsymbol{\theta}_k^\top \mathbf{x})}{\sum_{j=1}^{g} \exp(\boldsymbol{\theta}_j^\top \mathbf{x})}, k \in \{1, ..., g\}$$

a) Show that logistic and softmax regression are equivalent for $g = 2$.

Solution

As we would expect, the two formulations are equivalent (up to reparameterization). In order to see this, consider the softmax function components for both classes:

$$\pi_1(\mathbf{x}|\boldsymbol{\theta}) = \frac{\exp(\boldsymbol{\theta}_1^\top \mathbf{x})}{\exp(\boldsymbol{\theta}_1^\top \mathbf{x}) + \exp(\boldsymbol{\theta}_2^\top \mathbf{x})}$$

$$\pi_2(\mathbf{x}|\boldsymbol{\theta}) = \frac{\exp(\boldsymbol{\theta}_2^\top \mathbf{x})}{\exp(\boldsymbol{\theta}_1^\top \mathbf{x}) + \exp(\boldsymbol{\theta}_2^\top \mathbf{x})}$$

Since we know that $\pi_1(\mathbf{x}|\boldsymbol{\theta}) + \pi_2(\mathbf{x}|\boldsymbol{\theta}) = 1$, it is sufficient to compute one of the two scoring functions. Let's pick $\pi_1(\mathbf{x}|\boldsymbol{\theta})$ and relate it to the logistic function:

$$\pi_1(\mathbf{x}|\boldsymbol{\theta}) = \frac{1}{1 + \exp(\boldsymbol{\theta}_2^\top \mathbf{x} - \boldsymbol{\theta}_1^\top \mathbf{x})} = \frac{1}{1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x})}$$

where $\boldsymbol{\theta} := \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2$. Thus, we obtain the binary-case logistic function, reflecting that we only need one scoring function (and thus one set of parameters $\boldsymbol{\theta}$ rather than two $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2$).

b) Recall that in the binary case we predict $\hat{y} = 1$ iff $\pi(\mathbf{x}) \geq \alpha$ for a threshold $\alpha \in (0, 1)$. Show that the resulting decision boundary is a (linear) hyperplane.

Hint

Derive the value of $\boldsymbol{\theta}^\top \mathbf{x}$ (depending on α) starting from which you predict $\hat{y} = 1$ rather than $\hat{y} = 0$.

Solution

We evaluate

$$\begin{aligned}
\pi(\mathbf{x}) = \frac{1}{1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x})} &= \alpha \\
\Leftrightarrow \; 1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x}) &= \frac{1}{\alpha} \\
\Leftrightarrow \; \exp(-\boldsymbol{\theta}^\top \mathbf{x}) &= \frac{1}{\alpha} - 1 \\
\Leftrightarrow \; -\boldsymbol{\theta}^\top \mathbf{x} &= \log\left(\frac{1}{\alpha} - 1\right) \\
\Leftrightarrow \; \boldsymbol{\theta}^\top \mathbf{x} &= -\log\left(\frac{1}{\alpha} - 1\right).
\end{aligned}$$

$\boldsymbol{\theta}^\top \mathbf{x} = -\log\left(\frac{1}{\alpha} - 1\right)$ is the equation of a linear hyperplane, consisting of all points $\mathbf{x}$ whose score $\boldsymbol{\theta}^\top \mathbf{x}$ equals $-\log\left(\frac{1}{\alpha} - 1\right)$. Since $\pi(\mathbf{x})$ is increasing in $\boldsymbol{\theta}^\top \mathbf{x}$, we set $\hat{y} = 1$ for all points on or above this hyperplane, i.e. with $\boldsymbol{\theta}^\top \mathbf{x} \geq -\log\left(\frac{1}{\alpha} - 1\right)$.

Exercise 2: Hyperplanes

Learning goals

1. Understand that hyperplanes bisect the space with a linear boundary
2. Get a feeling for coefficients in hyperplane equations
3. See how the classification threshold α shifts the logistic decision boundary

Linear classifiers like logistic regression learn a decision boundary that takes the form of a (linear) hyperplane. Hyperplanes are defined by equations $\boldsymbol{\theta}^\top \mathbf{x} = b$ with coefficients $\boldsymbol{\theta}$ and a scalar $b \in \mathbb{R}$.

a) In order to see that such expressions actually describe hyperplanes, consider $\boldsymbol{\theta}^\top \mathbf{x} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0$. Sketch the hyperplanes given by the following coefficients and explain the difference between the parameterizations:

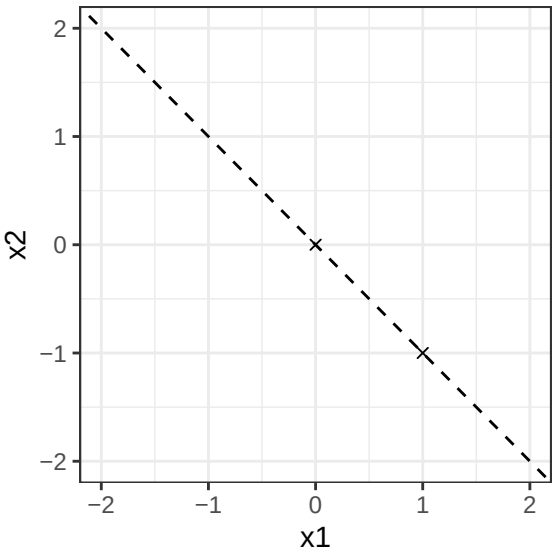
- $\theta_0 = 0, \theta_1 = \theta_2 = 1$
- $\theta_0 = 1, \theta_1 = \theta_2 = 1$
- $\theta_0 = 0, \theta_1 = 1, \theta_2 = 2$

Solution

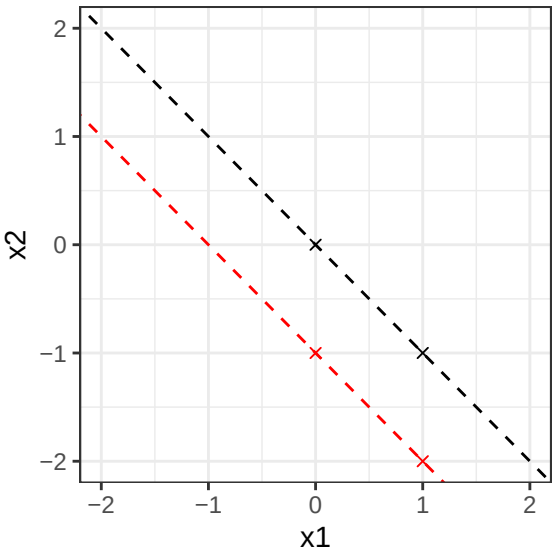
A hyperplane in 2D is just a line. We know that two points are sufficient to describe a line, so all we need to do is pick two points fulfilling the hyperplane equation.

- $\theta_0 = 0, \theta_1 = \theta_2 = 1 \rightsquigarrow$ e.g., (0, 0) and (1, -1). Sketch it:

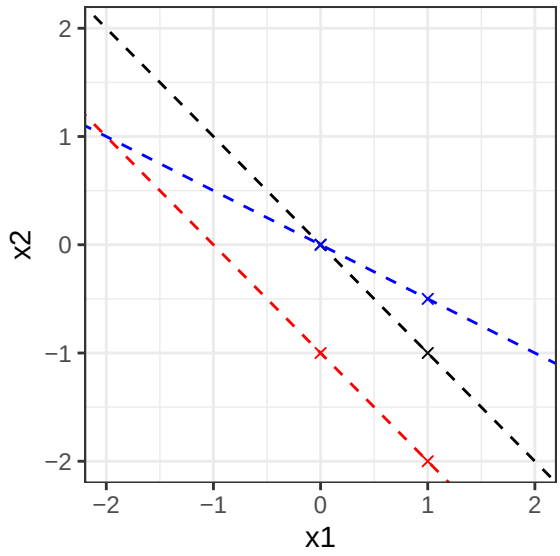
```
Warning: Paket 'ggplot2' wurde unter R Version 4.5.2 erstellt
```



- $\theta_0 = 1, \theta_1 = \theta_2 = 1 \rightsquigarrow$ e.g., (0, -1) and (1, -2). The change in θ_0 promotes a horizontal shift:



- $\theta_0 = 0, \theta_1 = 1, \theta_2 = 2 \rightsquigarrow$ e.g., (0, 0) and (1, -0.5). The change in θ_2 pivots the line around the intercept:



We see that a hyperplane is defined by the points that lie directly on it and thus fulfill the hyperplane equation.

b) Beyond its orientation, a linear classifier also has an *offset* that determines where exactly the boundary sits. Recall that a logistic regression model estimates $\pi(\mathbf{x}) = \frac{1}{1+\exp(-\boldsymbol{\theta}^\top \mathbf{x})}$ and predicts $\hat{y} = 1$ iff $\pi(\mathbf{x}) \geq \alpha$ for a threshold $\alpha \in (0, 1)$. Its decision boundary is therefore the hyperplane

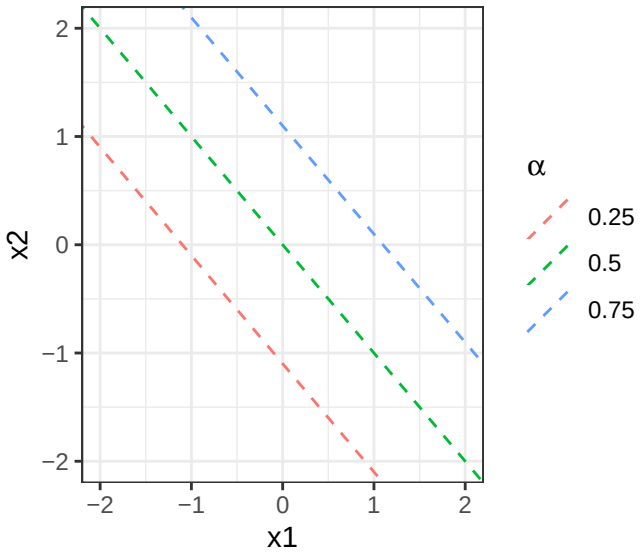
$$\boldsymbol{\theta}^\top \mathbf{x} = -\log\left(\frac{1}{\alpha} - 1\right) =: b(\alpha).$$

Consider the fixed model $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2)^\top = (0, 1, 1)^\top$. Compute the offset $b(\alpha)$ for $\alpha \in \{0.25, 0.5, 0.75\}$, sketch the three resulting decision boundaries in a single (x_1, x_2) plot, and describe how the boundary – and the region predicted as class 1 – changes as α grows. Does the orientation of the boundary change?

Solution

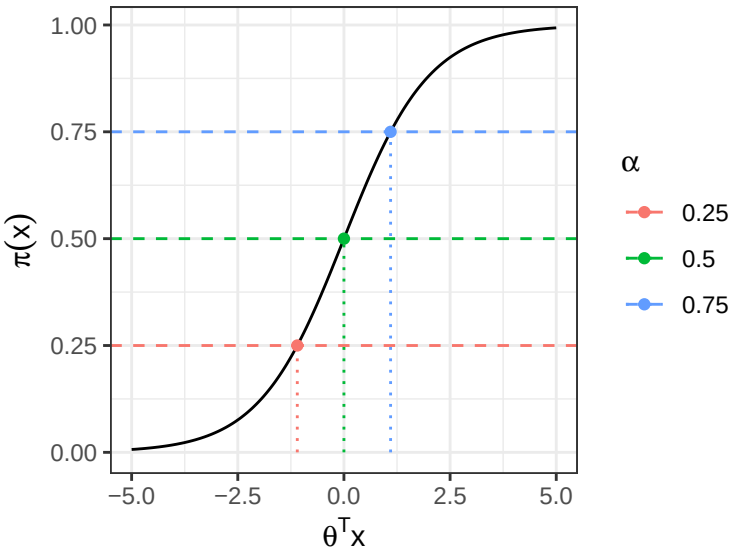
For this model, $\boldsymbol{\theta}^\top \mathbf{x} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 = x_1 + x_2$, so every decision boundary is a line $x_1 + x_2 = b(\alpha)$ with slope -1 and intercept $b(\alpha)$:

- $\alpha = 0.25$: $b(\alpha) = -\log\left(\frac{1}{0.25} - 1\right) = -\log 3 \approx -1.10$
- $\alpha = 0.5$: $b(\alpha) = -\log\left(\frac{1}{0.5} - 1\right) = -\log 1 = 0$
- $\alpha = 0.75$: $b(\alpha) = -\log\left(\frac{1}{0.75} - 1\right) = -\log \frac{1}{3} = \log 3 \approx 1.10$



As α increases, the offset $b(\alpha)$ increases and the boundary shifts *parallel* toward the upper right (larger $\boldsymbol{\theta}^\top \mathbf{x}$). Since we predict $\hat{y} = 1$ wherever $\boldsymbol{\theta}^\top \mathbf{x} \geq b(\alpha)$, the region classified as class 1 (the upper-right halfplane) shrinks: a higher threshold makes the model more conservative about predicting class 1. The orientation does **not** change – the normal vector $\boldsymbol{\theta} = (1, 1)^\top$ stays fixed, so all three boundaries are parallel. This mirrors part a), where changing the intercept θ_0 produced a parallel shift of the hyperplane.

Crucially, the fitted model itself – the logistic (S-)curve $\pi(\mathbf{x}) = \frac{1}{1+\exp(-\boldsymbol{\theta}^\top \mathbf{x})}$ – does **not** change when we vary α; it depends on $\boldsymbol{\theta}$ alone. Varying α only moves the *horizontal* cut-off through this fixed curve:



We read off the score $b(\alpha)$ at which the curve reaches height $\pi(\mathbf{x}) = \alpha$ (dashed lines) and classify every point with $\boldsymbol{\theta}^\top \mathbf{x} \geq b(\alpha)$ as class 1. Thus α is a *post-hoc decision threshold* applied to fixed predicted probabilities, not a re-fitting of the model. The three cut-off scores $b(\alpha) \in \{-1.10, 0, 1.10\}$ (dotted verticals) are exactly the intercepts of the three parallel decision boundaries in the (x_1, x_2) plot above.

Exercise 3: Decision Boundaries & Thresholds in Logistic Regression

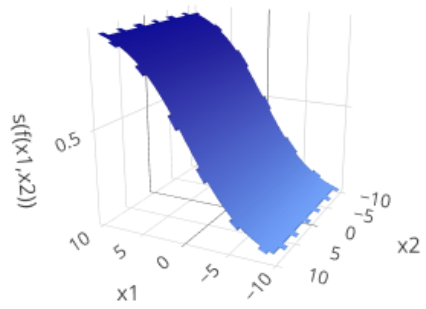
Learning goals

1. Understand that logistic regression finds a linear decision boundary
2. Get a feeling for how the parameters $\boldsymbol{\theta}$ shape predicted probabilities and model confidence
3. Understand the role of the decision threshold α

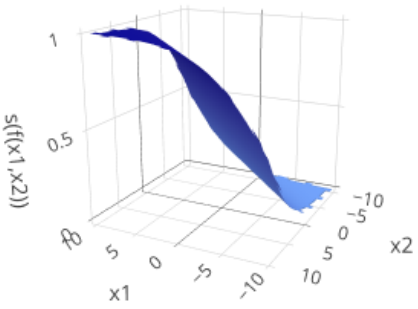
Recall from Exercise 2 that a logistic regression model estimates the probability $p(y = 1|\mathbf{x}, \boldsymbol{\theta}) = \pi(\mathbf{x}|\boldsymbol{\theta})$ and assigns $\hat{y} = 1$ iff $\pi(\mathbf{x}|\boldsymbol{\theta}) \geq \alpha$ for a threshold $\alpha \in (0, 1)$. Whereas Exercise 2 b) varied the threshold for a fixed model, we now study how the parameters $\boldsymbol{\theta}$ themselves shape the predicted probabilities.

a) Below you see the logistic function for a binary classification problem with two input features for different values $\boldsymbol{\theta}^\top = (\theta_1, \theta_2)^\top$ (plots 1-3) as well as α (plot 4). What can you deduce for the values of θ_1, θ_2, and α? What are the implications for classification in the different scenarios? In particular, which of the plots (1)–(3) shows the most confident model?

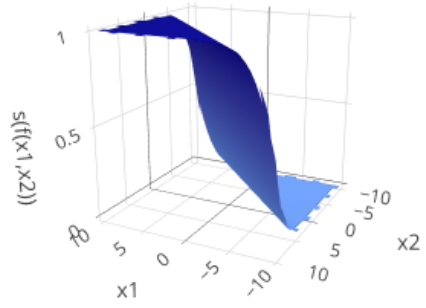
Plot (1)



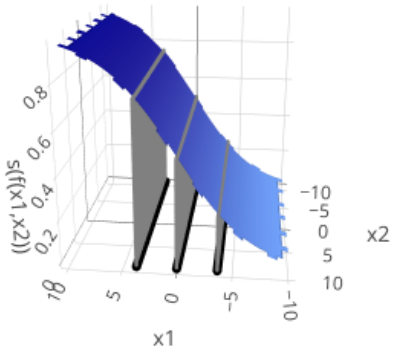
Plot (2)



Plot (3)



Plot (4)



Solution

We observe

- in plot (1): the logistic function runs parallel to the x_2 axis, so it is the same for every value of x_2. In other words, x_2 does not contribute anything to the class discrimination and its associated parameter θ_2 is equal to 0.
- in plot (2): both dimensions affect the logistic function – to equal degree in this case, meaning x_1 and x_2 are equally important. If θ_1 were larger than θ_2 or vice versa the hypersurface would be more tilted towards the respective axis. Furthermore, due to θ_1 and θ_2 being positive, $\pi(\mathbf{x})$ increases with higher values for x_1 and x_2.
- in plot (3): this is the same situation as in plot (2) but the logistic function is steeper, which is due to θ_1, θ_2 having larger absolute values. We therefore get a sharper separation between classes (fewer predicted probability values close to 0.5, so we are overall more confident in our decision). As in plot (2), the increasing probability of $\hat{y} = 1$ for higher values of x_1 and x_2 indicates positive values for θ_1 and θ_2.
- in plot (4): this is the same situation as in plot (1). The different values for α represent different thresholds: a high value (leftmost line) means we only assign class 1 if the estimated class-1 probability is large. Conversely, a low value (rightmost line) signifies we are ready to predict class 1 at a low threshold – in effect, this is the same as the previous scenario, only the class labels are flipped. The mid line corresponds to the common case $\alpha = 0.5$ where we assign class 1 as soon as the predicted probability is more than 50%.

In summary, **plot (3)** shows the most confident model: its steeper logistic surface pushes predicted probabilities closer to 0 or 1, yielding the sharpest class separation.

b) In Exercise 2 b) we simply read off the offset $b(0.5) = 0$ for the common threshold $\alpha = 0.5$. Derive this special case directly, starting from $\pi(\mathbf{x}) = 0.5$.

Solution

At the decision boundary the model is exactly indifferent between the two classes, i.e., $\pi(\mathbf{x}) = \alpha = 0.5$:

$$\begin{aligned} \pi(\mathbf{x}) = \frac{1}{1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x})} = 0.5 &\Leftrightarrow 1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x}) = 2 \\ &\Leftrightarrow \exp(-\boldsymbol{\theta}^\top \mathbf{x}) = 1 \\ &\Leftrightarrow \boldsymbol{\theta}^\top \mathbf{x} = 0. \end{aligned}$$

The 0.5 threshold therefore leads to the coordinate hyperplane $\boldsymbol{\theta}^\top \mathbf{x} = 0$ and divides the input space into the positive "1" halfspace where $\boldsymbol{\theta}^\top \mathbf{x} \geq 0$ and the "0" halfspace where $\boldsymbol{\theta}^\top \mathbf{x} < 0$.

c) Explain when it might be sensible to set α to 0.5.

Solution

When the threshold $\alpha = 0.5$ is chosen, the losses of misclassified observations, i.e., $L(\hat{y} = 0 \mid y = 1)$ and $L(\hat{y} = 1 \mid y = 0)$, are treated equally, which is often the intuitive thing to do. It means $\alpha = 0.5$ is a sensible threshold if we do not wish to avoid one type of misclassification more than the other. If, however, we need to be cautious to only predict class 1 if we are very confident (for example, when the decision triggers a costly therapy), it would make sense to set the threshold considerably higher.

Exercise 4: Thresholds and predictions in practice

The whole exercise is only for lecture group B!

Learning goals

1. Compute predictions of a logistic model by hand
2. Understand how the threshold α trades off the two error types

Consider a fitted binary logistic regression model with parameter vector $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2)^\top = (-1, 2, 1)^\top$, so that

$$\pi(\mathbf{x}) = \frac{1}{1 + \exp\left(-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)\right)}.$$

Recall that we predict $\hat{y} = 1$ iff $\pi(\mathbf{x}) \geq \alpha$.

a) Compute the predicted probability $\pi(\mathbf{x})$ for the following observations and classify them using the default threshold $\alpha = 0.5$:

- $\mathbf{x}^{(1)} = (x_1, x_2)^\top = (1, 1)^\top$
- $\mathbf{x}^{(2)} = (1, -1)^\top$
- $\mathbf{x}^{(3)} = (0, 0)^\top$

Solution

We plug each observation into $\boldsymbol{\theta}^\top \mathbf{x} = \theta_0 + \theta_1 x_1 + \theta_2 x_2$ and apply the logistic function:

- $\mathbf{x}^{(1)}$: $\boldsymbol{\theta}^\top \mathbf{x} = -1 + 2 \cdot 1 + 1 \cdot 1 = 2 \;\Rightarrow\; \pi(\mathbf{x}) = \frac{1}{1+\exp(-2)} \approx 0.88 \;\Rightarrow\; \hat{y} = 1$

- $\mathbf{x}^{(2)}$: $\boldsymbol{\theta}^\top \mathbf{x} = -1 + 2 \cdot 1 + 1 \cdot (-1) = 0 \;\Rightarrow\; \pi(\mathbf{x}) = \frac{1}{1+\exp(0)} = 0.5 \;\Rightarrow\; \hat{y} = 1$ (exactly on the boundary, where $\pi(\mathbf{x}) = \alpha$)
- $\mathbf{x}^{(3)}$: $\boldsymbol{\theta}^\top \mathbf{x} = -1 + 2 \cdot 0 + 1 \cdot 0 = -1 \;\Rightarrow\; \pi(\mathbf{x}) = \frac{1}{1+\exp(1)} \approx 0.27 \;\Rightarrow\; \hat{y} = 0$

Note that $\mathbf{x}^{(2)}$ lies exactly on the $\alpha = 0.5$ decision boundary $\boldsymbol{\theta}^\top \mathbf{x} = 0$ derived in Exercise 3 b).

b) For observation $\mathbf{x}^{(1)}$, from which threshold α onwards would the model predict $\hat{y} = 0$ instead?

Solution

Since $\hat{y} = 1$ iff $\pi(\mathbf{x}) \geq \alpha$, the prediction for $\mathbf{x}^{(1)}$ switches to $\hat{y} = 0$ as soon as $\alpha > \pi(\mathbf{x}^{(1)}) \approx 0.88$. In other words, we would only predict class 0 for this observation if we demanded a class-1 probability of more than about 88%.

c) The model is used in a hospital to decide whether to start an expensive and invasive therapy ($\hat{y} = 1$). Would you rather choose a high or a low threshold α? Which type of error (false positive / false negative) does your choice avoid, and at what cost?

Solution

If the therapy itself is costly and invasive, we want to be sure a patient really needs it before starting it, i.e., only predict $\hat{y} = 1$ when we are very confident. This calls for a **high** threshold α: it reduces **false positives** (healthy patients undergoing an unnecessary, harmful therapy), at the price of more **false negatives** (patients who would benefit are not treated).

The "right" choice therefore depends on the relative costs of the two error types. If missing a case were far more dangerous than an unnecessary treatment (e.g., a life-threatening but easily treatable disease), we would instead prefer a **low** α to avoid false negatives.